Heat Transfer Computer Project 1

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1 Problem 1

1.1 Governing Equation

The temperature distribution in the solar panel is governed by the steady-state heat conduction equation:

$$\nabla \cdot (k \nabla T) = 0 \tag{1}$$

The computational domain is discretized into a uniform grid with:

$$\Delta x = \frac{0.45}{225}, \quad \Delta y = \frac{0.06}{30} \tag{2}$$

A finite-difference formulation based on energy balance is applied to each node.

1.2 Boundary Conditions

The following boundary conditions are applied to the computational domain:

1.2.1 Top Surface (Radiative Heat Flux)

The top surface of the PV panel is subjected to a uniform heat flux:

$$-k \frac{\partial T}{\partial y} = q'' \tag{3}$$

1.2.2 Channel Surfaces

At the surfaces of the cooling channels, convection occurs between the solid and the fluid:

$$-k \frac{\partial T}{\partial n} = h(T - T_\infty) \tag{4}$$

where h is the convective heat transfer coefficient and T_∞ is the fluid temperature.

1.2.3 Insulated Boundaries

The left, right, and bottom boundaries are insulated, thus:

$$\frac{\partial T}{\partial n} = 0 \tag{5}$$

1.2.4 Interface Between PV and Aluminum

At the interface between the PV layer and the aluminum layer, both temperature continuity and heat flux continuity must be satisfied:

$$T_{PV} = T_{Al} \tag{6}$$

$$k_{PV}\frac{\partial T}{\partial y} = k_{Al}\frac{\partial T}{\partial y} \tag{7}$$

1.3 Node Equations

1.3.1 Interior Nodes

For interior nodes:

$$T_{m+1,n} + T_{m-1,n} + T_{m,n+1} + T_{m,n-1} - 4T_{m,n} = 0 \tag{8}$$

This applies to both PV and aluminum regions.

1.3.2 Top Surface Nodes (Heat Flux Boundary)

The top surface is subjected to a uniform heat flux q''.

(a) Interior nodes on top surface:

$$(2T_{m-1,n} + T_{m,n+1} + T_{m,n-1}) + \frac{2q''\Delta x}{k} - 4T_{m,n} = 0 \tag{9}$$

(b) Top-left corner:

$$(2T_{m+1,n} + T_{m,n-1}) + \frac{2q''\Delta x}{k} - 3T_{m,n} = 0 \tag{10}$$

(c) Top-right corner:

$$(2T_{m-1,n} + T_{m,n-1}) + \frac{2q''\Delta x}{k} - 3T_{m,n} = 0 \tag{11}$$

1.3.3 Insulated Boundaries

For insulated boundaries, $\frac{\partial T}{\partial n} = 0$.

(a) Left boundary:

$$2T_{m+1,n} + T_{m,n+1} + T_{m,n-1} - 4T_{m,n} = 0 \tag{12}$$

(b) Right boundary:

$$2T_{m-1,n} + T_{m,n+1} + T_{m,n-1} - 4T_{m,n} = 0 \tag{13}$$

(c) Bottom boundary:

$$T_{m+1,n} + T_{m-1,n} + 2T_{m,n+1} - 4T_{m,n} = 0 \tag{14}$$

1.3.4 Channel Nodes (Convective Boundary)

At the channel surfaces, convection is applied:

$$-k\frac{\partial T}{\partial n} = h(T - T_\infty) \tag{15}$$

(a) Channel top surface:

$$(2T_{m,n-1} + T_{m+1,n} + T_{m-1,n}) + \frac{2h\Delta x}{k}T_\infty - 2\left(\frac{h\Delta x}{k} + 2\right)T_{m,n} = 0 \tag{16}$$

(b) Channel bottom surface:

$$(2T_{m,n+1} + T_{m+1,n} + T_{m-1,n}) + \frac{2h\Delta x}{k}T_\infty - 2\left(\frac{h\Delta x}{k} + 2\right)T_{m,n} = 0 \tag{17}$$

(c) Channel left surface:

$$(2T_{m+1,n} + T_{m,n+1} + T_{m,n-1}) + \frac{2h\Delta x}{k}T_\infty - 2\left(\frac{h\Delta x}{k} + 2\right)T_{m,n} = 0 \tag{18}$$

(d) Channel right surface:

$$(2T_{m-1,n} + T_{m,n+1} + T_{m,n-1}) + \frac{2h\Delta x}{k}T_\infty - 2\left(\frac{h\Delta x}{k} + 2\right)T_{m,n} = 0 \tag{19}$$

(e) Channel top-left corner:

$$2(T_{m+1,n} + T_{m,n-1}) + (T_{m-1,n} + T_{m,n+1}) + 2\frac{h\Delta x}{k}T_\infty - 2\left(3 + \frac{h\Delta x}{k}\right)T_{m,n} = 0 \tag{20}$$

(f) Channel top-right corner:

$$2(T_{m-1,n} + T_{m,n-1}) + (T_{m+1,n} + T_{m,n+1}) + 2\frac{h\Delta x}{k}T_\infty - 2\left(3 + \frac{h\Delta x}{k}\right)T_{m,n} = 0 \tag{21}$$

(g) Channel bottom-left corner:

$$2(T_{m+1,n} + T_{m,n+1}) + (T_{m-1,n} + T_{m,n-1}) + 2\frac{h\Delta x}{k}T_\infty - 2\left(3 + \frac{h\Delta x}{k}\right)T_{m,n} = 0 \tag{22}$$

(h) Channel bottom-right corner:

$$2(T_{m-1,n} + T_{m,n+1}) + (T_{m+1,n} + T_{m,n-1}) + 2\frac{h\Delta x}{k}T_\infty - 2\left(3 + \frac{h\Delta x}{k}\right)T_{m,n} = 0 \tag{23}$$

1.3.5 Interface Between PV and Aluminum

At the interface between the PV layer and the aluminum layer, both temperature continuity and heat flux continuity must be satisfied.

(a) Interior interface nodes From heat flux continuity:

$$k_{PV}(T_{m,n} - T_{m,n+1}) = k_{Al}(T_{m,n-1} - T_{m,n}) \tag{24}$$

Rearranging:

$$k_{PV}T_{m,n+1} + k_{Al}T_{m,n-1} - (k_{PV} + k_{Al})T_{m,n} = 0 \tag{25}$$

(b) Interface nodes adjacent to left insulated boundary At the left boundary, the insulation condition gives:

$$\frac{\partial T}{\partial x} = 0 \quad \Rightarrow \quad T_{m-1,n} = T_{m,n} \tag{26}$$

Thus, the interface equation remains:

$$k_{PV}(T_{m,n} - T_{m,n+1}) = k_{Al}(T_{m,n-1} - T_{m,n}) \tag{27}$$

(c) Interface nodes adjacent to right insulated boundary Similarly, for the right boundary:

$$\frac{\partial T}{\partial x} = 0 \quad \Rightarrow \quad T_{m+1,n} = T_{m,n} \tag{28}$$

Thus, the interface equation is:

$$k_{PV}(T_{m,n} - T_{m,n+1}) = k_{Al}(T_{m,n-1} - T_{m,n}) \tag{29}$$

2 Problem 2 and 3

For a water temperature of 20°C, the computed maximum temperature of the solar panel is 56.01°C, occurring at $(x, y) = (0.000, 0.000)$ m, which corresponds to the top-left corner of the PV layer. This is expected since the top surface is subjected to a uniform heat flux, and the corner experiences reduced lateral heat dissipation due to the insulated boundary.

The temperature distribution (Figure 1) shows that the PV layer reaches higher temperatures because of its low thermal conductivity ($k_{pv} = 1$ W/m·K), while the aluminum layer remains cooler due to its high conductivity ($k_{al} = 205$ W/m·K). The embedded cooling channels effectively reduce temperatures locally, creating cooler regions within the aluminum layer.

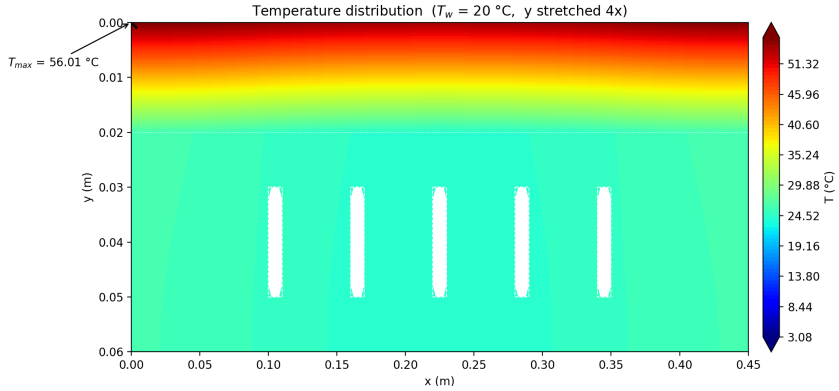

Figure 1: Temperature Distribution for Problem 2

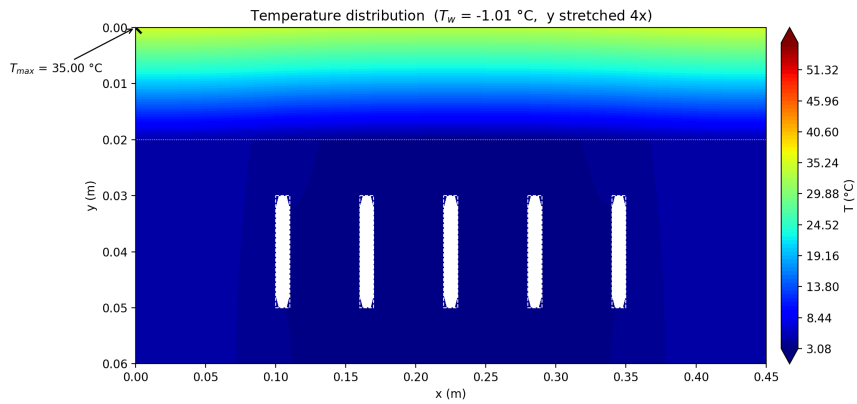

Figure 2: Temperature Distribution for Problem 3

To ensure that the panel temperature does not exceed 35°C, the maximum allowable water temperature is determined using the linear offset:

$$T_w^{max} = 35 - (T_{max} - T_w) = -1.008°\text{C}.$$

And the temperature distribution is shown in Figure 2. This result indicates that, under the given operating conditions, the cooling system must operate at sub-zero water temperatures to maintain safe panel operation.